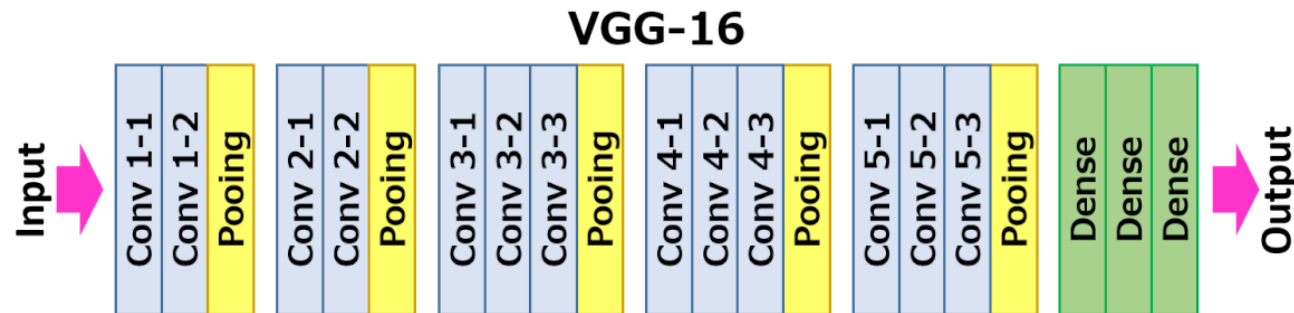


HW# 7

Deep Learning for Computer Vision (EN525.733.8VL)

Due date: April 9th, 2026

- Design a Siamese network as an object verifier (matcher) using a VGGNet-16 architecture as your network.
- Use CIFAR-10 data set and train your Siamese network. Use the output of the last fully connected as the feature embedding vectors (4096 feature values) by discarding the Softmax layer. Use the training and testing dataset specified in the CIFAR-10 database. (<https://www.cs.toronto.edu/~kriz/cifar.html>)
- Report verification performance of your Siamese network for each class. Plot the learning curve for the training, validation + test datasets. (remember you need to create **Genuine and imposter pairs for training and testing**, make sure the batch size has equal genuine and imposter pairs)
- See my notes + next page, it describes the contrastive loss function that needs to be minimized, play around with the margin value " m " (i.e., 1, 1.5, or 2).
- Please note you can get rid of the last two Pooling layers so that you can control the size of the feature maps.



Siamese Contrastive Loss Function

- Define an L2-norm distance (Euclidean):

$$D_W(\mathbf{x}_i, \mathbf{x}_j) = \|G_w(\mathbf{x}_i) - G_w(\mathbf{x}_j)\|_2$$

- For Genuine Pairs ($y=0$):

$$L_{gen}(D_W(\mathbf{x}_i, \mathbf{x}_j)) = \frac{1}{2} D_W(\mathbf{x}_i, \mathbf{x}_j)^2 = \frac{1}{2} \|G_w(\mathbf{x}_i) - G_w(\mathbf{x}_j)\|_2^2$$

- For Imposter pairs ($y=1$):

$$L_{imp}(D_W(\mathbf{x}_i, \mathbf{x}_j)) = \frac{1}{2} \max(0, (m - D_W(\mathbf{x}_i, \mathbf{x}_j)))^2$$

- m is a margin (anything inside the margin are genuine pairs and outside the margin are imposters).
- We have to minimize the total cost for N samples

$$\ell(W) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N L(W, (y, \mathbf{x}_i, \mathbf{x}_j))$$

$$L(W, (y, \mathbf{x}_i, \mathbf{x}_j)) = (1 - y)L_{gen}(D_W(\mathbf{x}_i, \mathbf{x}_j)) + yL_{imp}(D_W(\mathbf{x}_i, \mathbf{x}_j))$$

